UDP Log Viewer

Cross-platform UDP log viewer for ESP32 and other embedded systems, built with Python and PyQt5.

Documentation

Complete English documentation:

- docs/DOCUMENTATION_en.md
- docs/USER_GUIDE_en.md
- docs/RELEASE_0.16.0.md

Complete German documentation:

- docs/DOKUMENTATION_de.md

Windows-User

please use setup.exe to install this viewer

ToDo for MAC users

after installation this application is not signed and can't be started directly. You have to set this command: `xattr -dr com.apple.quarantine /Applications/UDPLogViewer.app`

Screenshots

Screenshot assets are stored under assets/screenshots.

Main Screen

Rule Configuration

Project Dialog

Plot Visualizer Tooltip

Logic Measurement

Current Scope

The current codebase includes:

- real-time UDP log reception
- filter, exclude, and highlight rules
- live session logging
- main-window `RESET` support to start a fresh log phase inside the same app session
- runtime project contexts for grouping logs and screenshots per test session

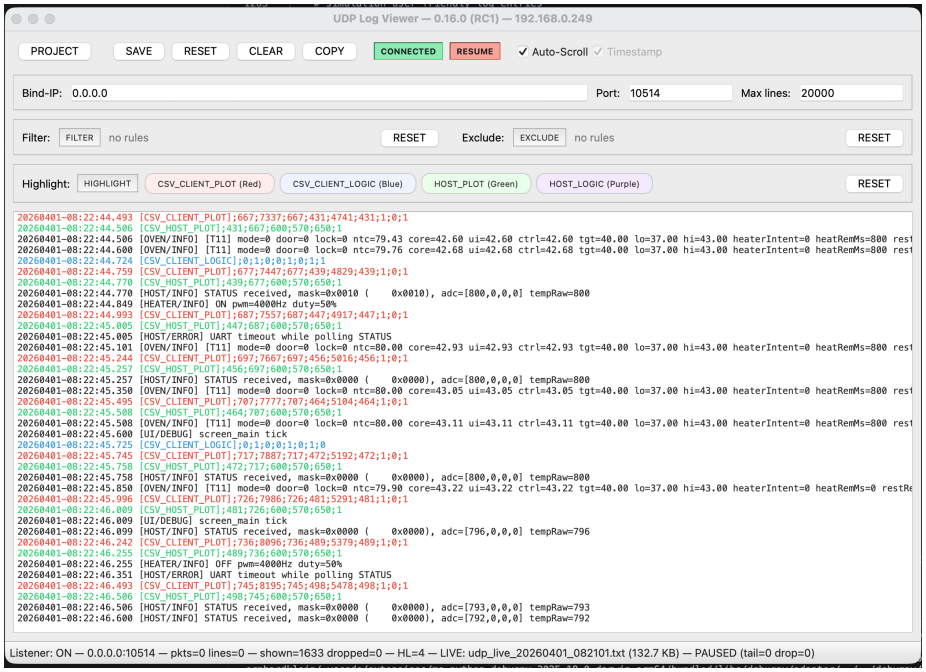

Figure 1: UDP Log Viewer main screen with active connection, reset workflow, and rule chips

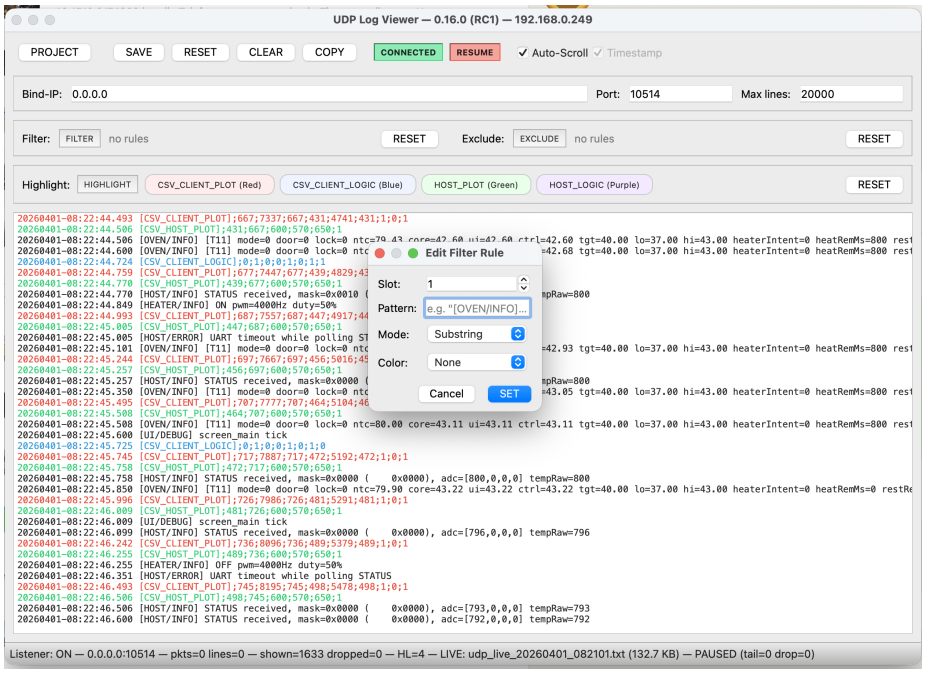

Figure 2: UDP Log Viewer filter-rule dialog on top of the main window

- project descriptions stored as `README_<projectname>.md` inside the project folder
- replay of saved log files
- built-in simulation for text, temperature, and logic traffic
- CSV-based data visualization
- logic-channel visualization
- edge and period measurement directly inside the logic graph
- runtime `Legend` toggle in plot and logic graph windows
- keyboard-driven screenshot/save shortcuts and explicit `TAB` navigation in the main and graph windows
- macOS and Windows packaging scripts

Run From Source

```
python -m venv venv
source venv/bin/activate
pip install -e .[dev]
udp-log-viewer
```

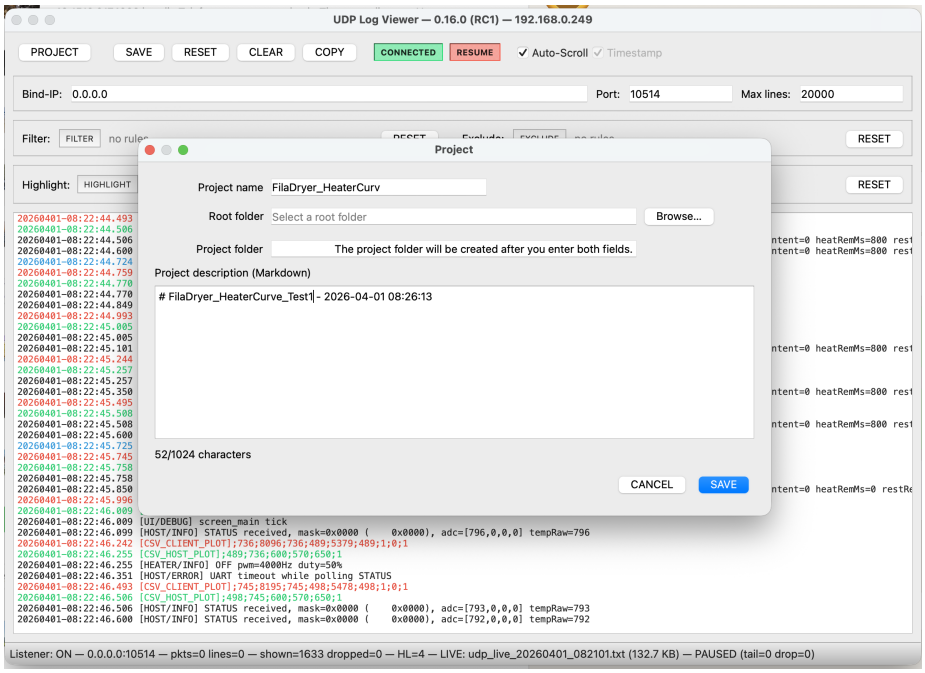

Figure 3: UDP Log Viewer project dialog with Markdown project description

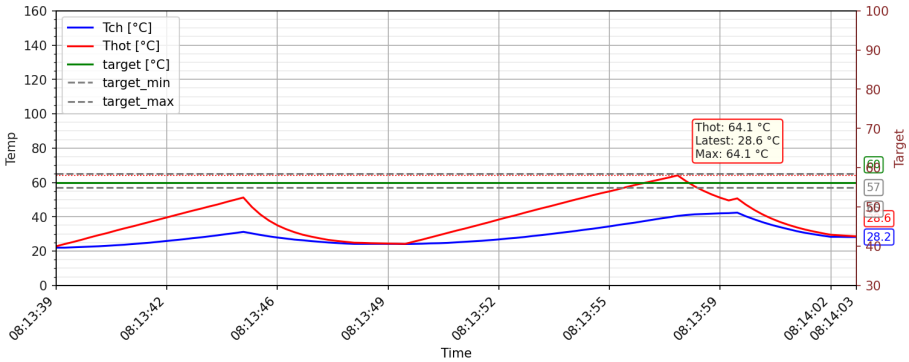

Figure 4: Plot visualizer with legend, target lines, and tooltip values

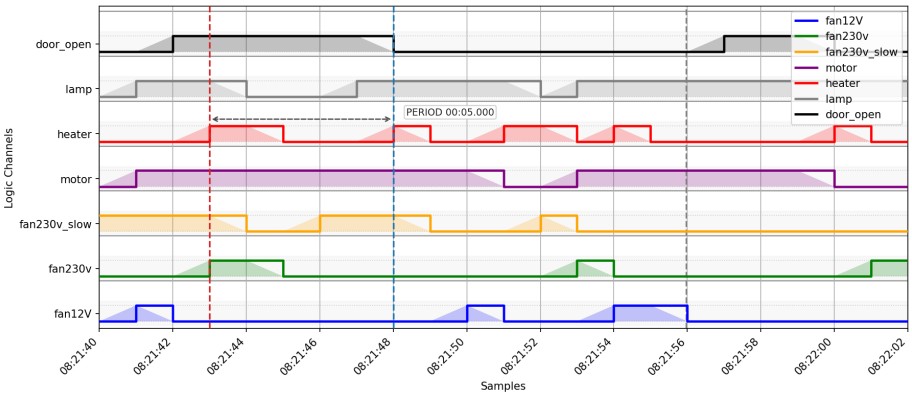

Figure 5: Logic visualizer period measurement with red and blue markers

Developer Bootstrap

```
./scripts/bootstrap_macos_linux.sh
```